

Statistical Analysis, Classification and Clustering on the UCI Wine Recognition Dataset

Data Science Project Report

Dataset:	UCI Wine Recognition Dataset
Source:	UCI Machine Learning Repository (archive.ics.uci.edu/ml/datasets/Wine)
Samples:	178
Features:	13 (all numerical)
Classes:	class_0, class_1, class_2 (3 wine cultivars)
Tools Used:	Python, pandas, NumPy, scikit-learn, matplotlib, seaborn

1. Dataset Description

The dataset used for this project is the **Wine Recognition Dataset** from the UCI Machine Learning Repository. It is the result of a chemical analysis of wines grown in the same region of Italy but derived from three different cultivars (grape varieties). The analysis determined the quantities of 13 chemical constituents found in each of the three types of wine.

The dataset contains **178 rows (samples)** and **13 numerical feature columns**, plus one target column denoting the class (cultivar) of each wine sample. All 13 features are continuous numeric measurements, which makes this dataset directly suitable for statistical analysis, classification, and clustering without any categorical encoding.

1.1 Feature Columns

#	Feature	Description
1	alcohol	Alcohol content (%)
2	malic_acid	Malic acid concentration
3	ash	Ash content
4	alcalinity_of_ash	Alkalinity of ash
5	magnesium	Magnesium content (mg)
6	total_phenols	Total phenolic compounds
7	flavanoids	Flavanoid content
8	nonflavanoid_phenols	Non-flavanoid phenols
9	proanthocyanins	Proanthocyanin content
10	color_intensity	Colour intensity
11	hue	Hue of the wine
12	od280/od315_of_diluted_wines	OD280/OD315 ratio (protein content proxy)
13	proline	Proline amino acid content

1.2 Class Labels

Class	Cultivar Name	Number of Samples	Proportion
class_0	Class 0	59	33.1%
class_1	Class 1	71	39.9%
class_2	Class 2	48	27.0%

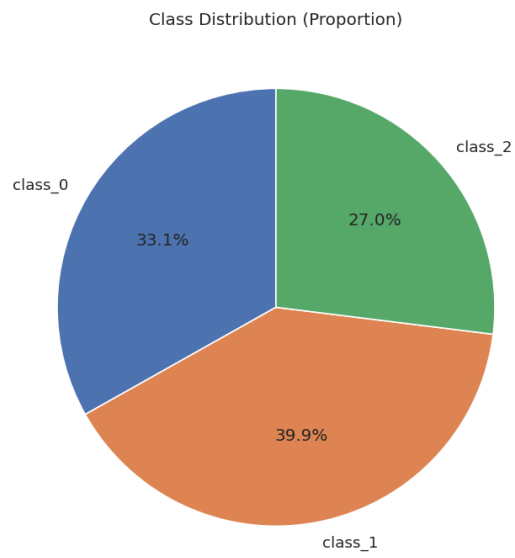


Figure 1.1: Proportion of samples belonging to each cultivar class.



Figure 1.2: Count of samples per class. The dataset is reasonably balanced across the three classes, which avoids the need for resampling techniques.

2. Data Preprocessing

2.1 Why Preprocessing Is Needed

Before applying any statistical or machine learning technique, the raw data was examined and prepared. The following preprocessing steps were carried out, along with the reasoning behind each:

- **Missing value check:** The dataset was checked for null/missing entries across all 13 feature columns. No missing values were found, so no imputation was required.
- **Type verification:** All 13 feature columns were confirmed to be numeric (float/int), satisfying the requirement that the data used for this analysis be numerical.
- **Feature scaling (Standardization):** The features are on very different scales – for example, *proline* ranges from 278 to 1680, while *hue* ranges from 0.48 to 1.71. Distance-based algorithms such as kNN, k-Means, DBSCAN, and SVM (with an RBF kernel) are sensitive to feature scale: a feature with a larger numeric range would dominate the distance calculation and bias the model. To address this, all features were standardized using **StandardScaler**, transforming each feature to have mean 0 and standard deviation 1.
- **Train/Test split:** For the classification task, the data was split into 70% training and 30% testing sets using stratified sampling, ensuring that the class proportions in both sets match the original dataset. This gives a fair, unbiased estimate of how well each classifier generalizes to unseen data.

2.2 Effect of Standardization

Before standardization, features like *proline* (mean $\approx$ 747, std $\approx$ 315) and *magnesium* (mean $\approx$ 100, std $\approx$ 14) had numeric ranges orders of magnitude larger than features like *hue* (mean $\approx$ 0.96, std $\approx$ 0.23) or *nonflavanoid_phenols* (mean $\approx$ 0.36). After standardization, every feature is on a common scale (mean 0, std 1), ensuring each of the 13 chemical measurements contributes proportionally to distance-based computations rather than being dominated by features measured in larger units.

2.3 Descriptive Statistics (Before Scaling)

The table below summarizes the mean, standard deviation, minimum and maximum values for each feature in the original (unscaled) dataset.

Feature	Mean	Std Dev	Min	Max
alcohol	13.00	0.81	11.03	14.83
malic_acid	2.34	1.12	0.74	5.80
ash	2.37	0.27	1.36	3.23
alcalinity_of_ash	19.49	3.34	10.60	30.00
magnesium	99.74	14.28	70.00	162.00
total_phenols	2.30	0.63	0.98	3.88
flavanoids	2.03	1.00	0.34	5.08
nonflavanoid_phenols	0.36	0.12	0.13	0.66
proanthocyanins	1.59	0.57	0.41	3.58
color_intensity	5.06	2.32	1.28	13.00
hue	0.96	0.23	0.48	1.71
od280/od315_of_diluted_wines	2.61	0.71	1.27	4.00
proline	746.89	314.91	278.00	1680.00

3. Task A — Preliminary Statistical Analysis & Inferences

3.1 Correlation Analysis

A Pearson correlation matrix was computed across all 13 features to identify linear relationships between chemical properties. The heatmap below visualizes these correlations, with dark red indicating strong positive correlation and dark blue indicating strong negative correlation.

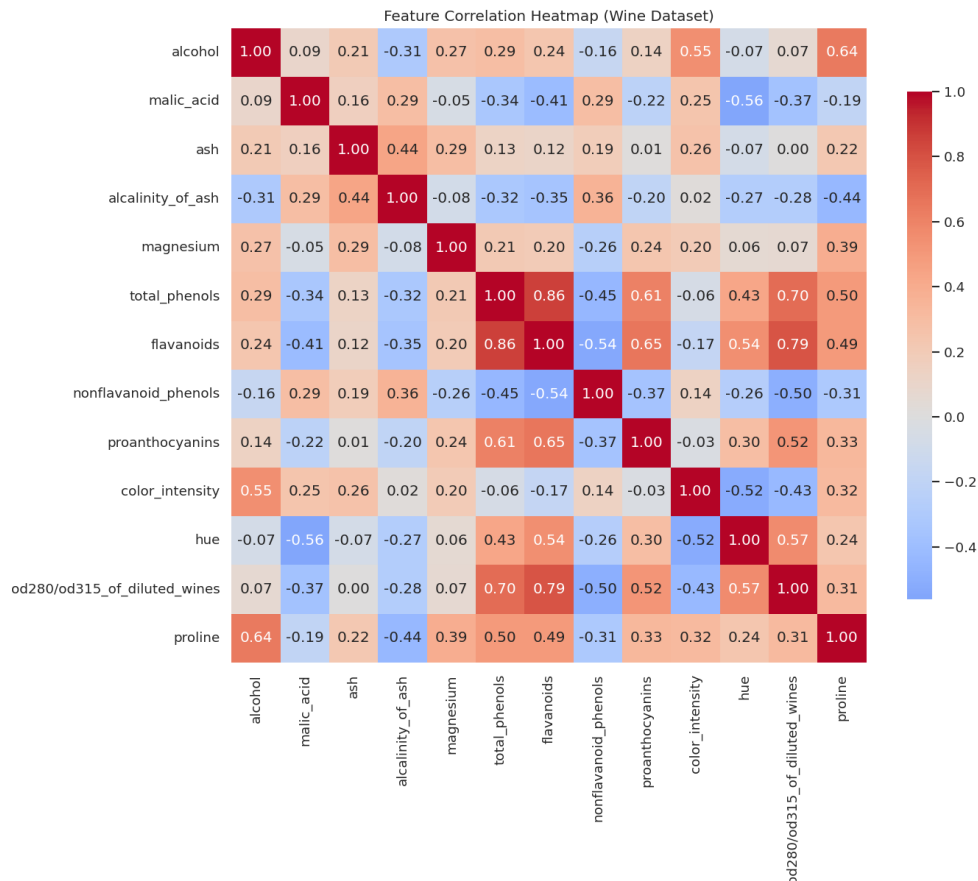


Figure 3.1: Correlation heatmap of all 13 chemical features.

Inferences from the correlation analysis:

- **total_phenols** and **flavanoids** show a very strong positive correlation ($r \approx 0.86$). This makes chemical sense – flavanoids are a sub-class of phenolic compounds, so wines high in total phenols also tend to be high in flavanoids.
- **flavanoids** and **od280/od315_of_diluted_wines** are also strongly correlated ($r \approx 0.79$), suggesting that the optical density measurement is heavily influenced by flavanoid content.
- **alcohol** and **proline** show a moderate positive correlation ($r \approx 0.64$), indicating wines with higher alcohol content tend to also have higher proline levels – both of which are useful discriminating features between the three cultivars, as seen in Section 3.2.
- **malic_acid** and **hue** show a moderate negative correlation ($r \approx -0.56$), meaning wines with higher malic acid content tend to have a lower hue value.
- **flavanoids** and **nonflavanoid_phenols** are negatively correlated ($r \approx -0.54$), which is expected since these represent two competing categories of the same broader phenol family.

3.2 Feature Distributions by Class

Boxplots of four key chemical features were generated, grouped by wine class, to identify which features best separate the three cultivars.

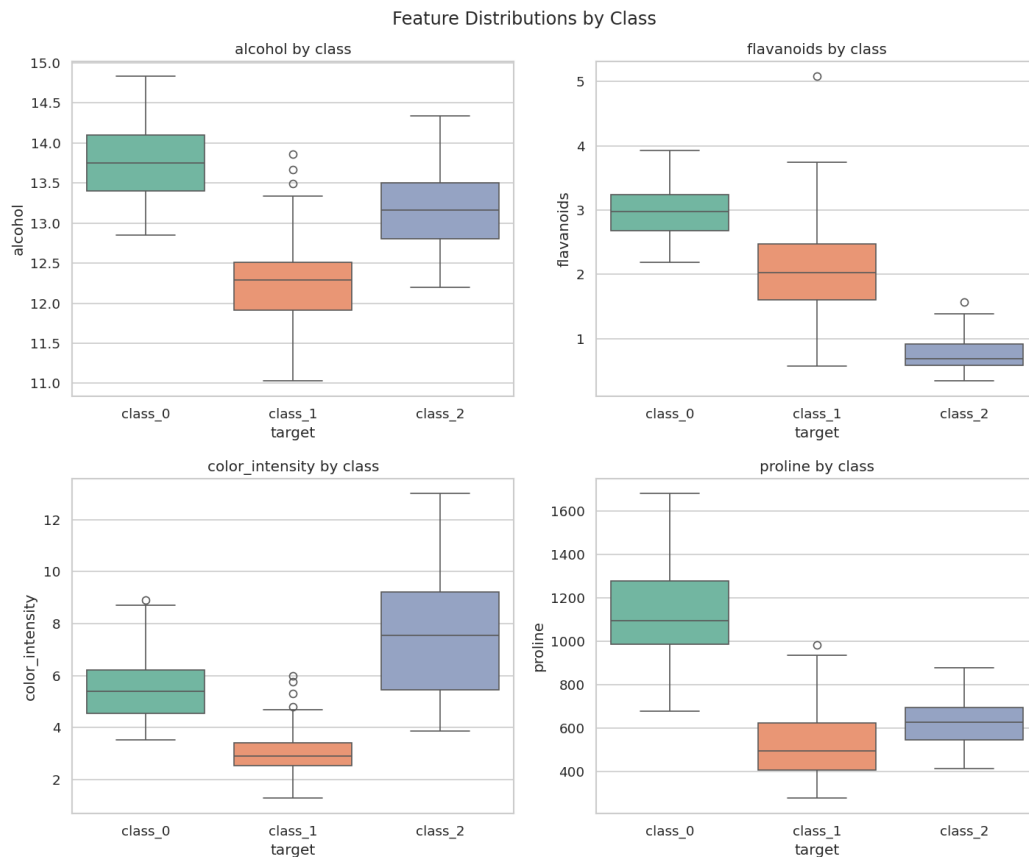


Figure 3.2: Boxplots of alcohol, flavanoids, color_intensity and proline by class.

Inferences:

- **Flavanoids** show the clearest class separation: class_0 and class_1 wines have noticeably higher flavanoid content than class_2, with relatively little overlap. This makes flavanoids a strong candidate feature for classification.
- **Proline** is markedly higher for class_0 wines compared to class_1 and class_2, making it another strong discriminating feature.
- **Color intensity** is highest for class_2 wines and lowest for class_1, providing a third useful axis of separation.
- **Alcohol** shows class_0 generally having the highest alcohol content, though with more overlap between classes than the other three features.
- Overall, these distributions explain why a classifier such as SVM achieves very high accuracy on this dataset (see Section 4) – the three classes are well-separated in the 13-dimensional feature space, even though no single feature perfectly separates all three on its own.

3.3 Feature Distributions (Histograms)

Histograms of all 13 features (across the entire dataset, all classes combined) were plotted to understand the overall shape of each feature's distribution.

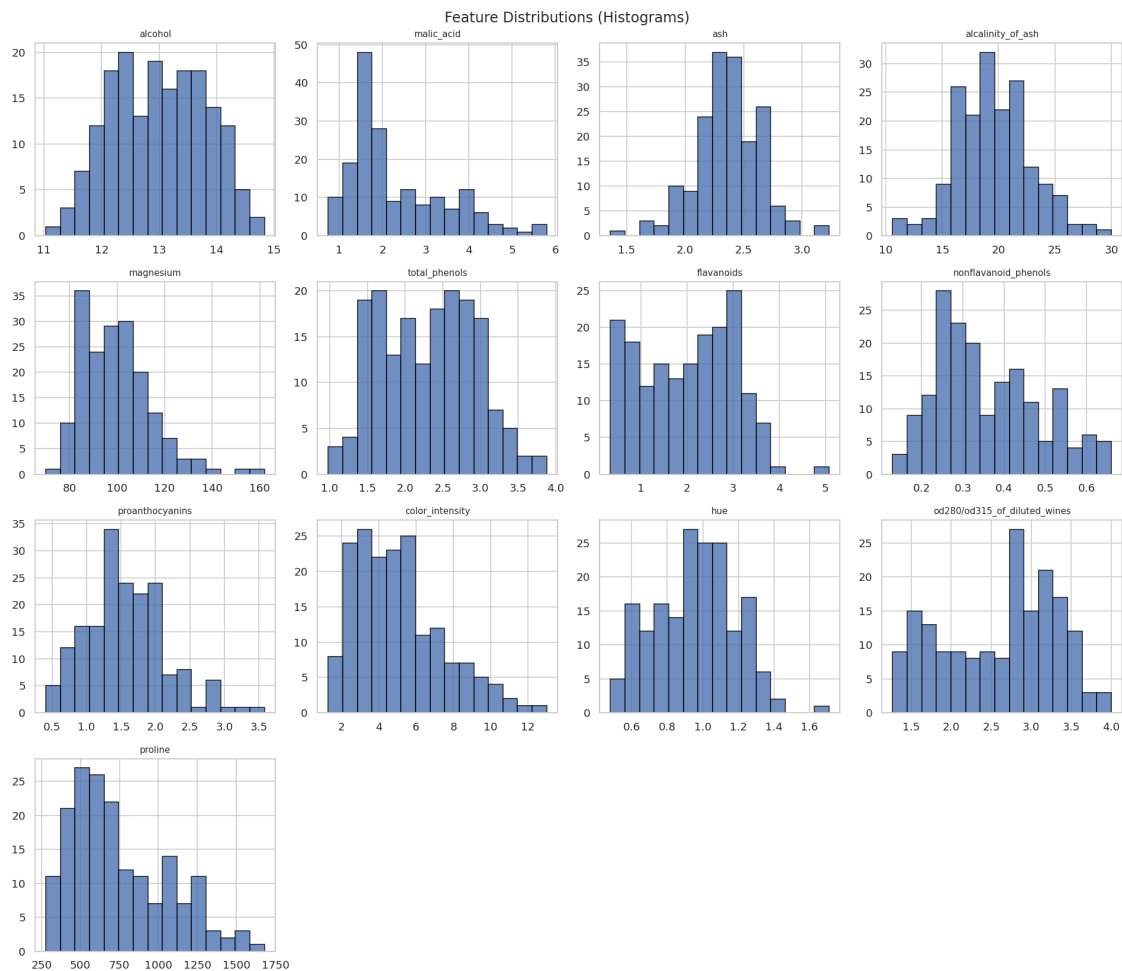


Figure 3.3: Histograms of all 13 features.

Inferences:

- Most features (alcohol, ash, alcalinity_of_ash, total_phenols, hue) approximate a roughly normal (bell-shaped) distribution, which is expected for natural chemical measurements across a sample population.
- **Proline** and **color_intensity** are right-skewed, with a long tail toward higher values – a small number of wines have unusually high proline or colour intensity compared to the bulk of the samples.
- **Malic_acid** is also right-skewed, with most wines clustered at lower values and a tail extending toward higher acid content.
- **Flavanoids** shows a roughly bimodal pattern, which is consistent with the boxplot finding that flavanoid content differs systematically between the wine classes – the two 'modes' likely correspond to the class_0/class_1 group (high flavanoids) versus class_2 (low flavanoids).

4. Task B — Classification (SVM and kNN)

4.1 Purpose and Methodology

The goal of this task is to predict the wine cultivar (class_0, class_1, or class_2) from its 13 chemical measurements, and to compare two classic classification algorithms: **Support Vector Machine (SVM)** with an RBF kernel, and **k-Nearest Neighbours (kNN)** with k=5. Both algorithms were trained on the standardized features using a 70/30 stratified train-test split, and evaluated using three metrics:

- **Accuracy** – the proportion of test samples correctly classified.
- **Cohen's Kappa** – a metric that compares observed accuracy to the accuracy expected by chance, correcting for class imbalance. A kappa close to 1 indicates the classifier's predictions agree with the true labels far more than random chance would suggest.
- **ROC-AUC (one-vs-rest)** – measures how well the classifier separates each class from the others across all probability thresholds, averaged over the three classes. A value of 1.0 indicates perfect separation.

4.2 Results Comparison

Metric	SVM (RBF kernel)	kNN (k=5)
Accuracy	0.9815	0.9444
Cohen's Kappa	0.9719	0.9167
ROC-AUC	1.0000	0.9953

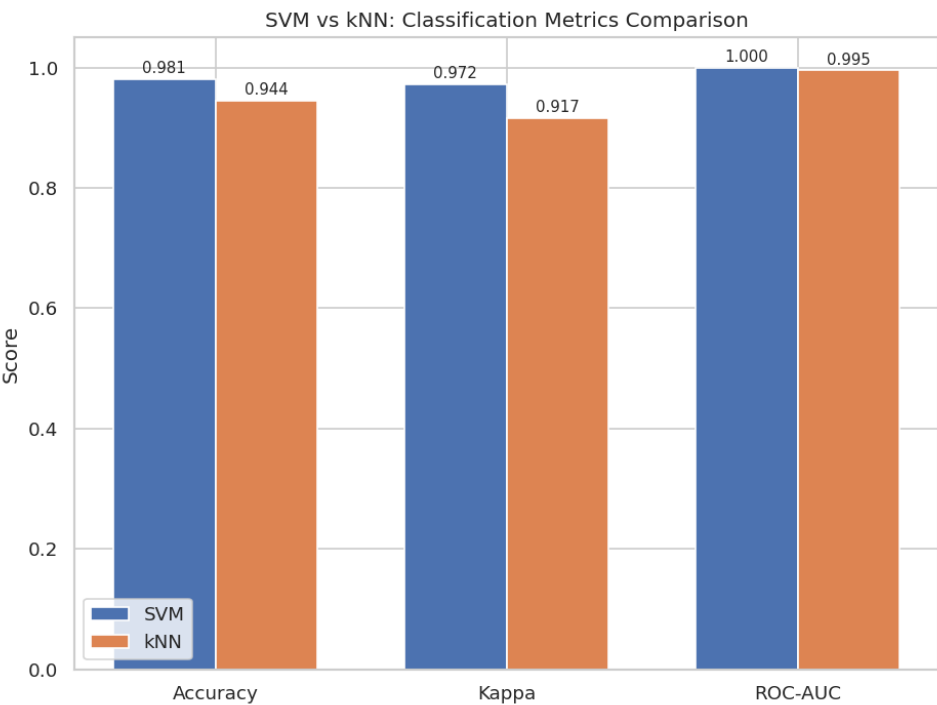


Figure 4.1: Bar chart comparing Accuracy, Kappa and ROC-AUC for SVM vs kNN.

4.3 Confusion Matrices

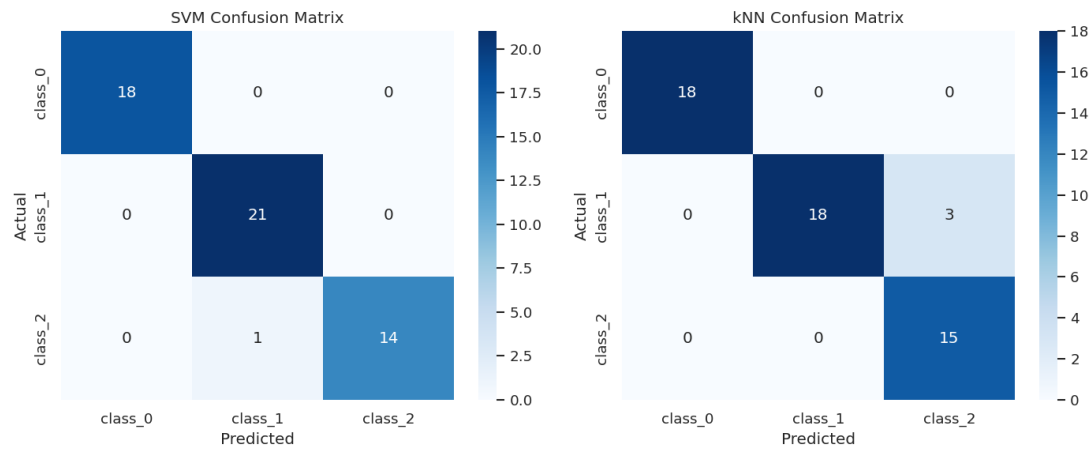


Figure 4.2: Confusion matrices on the 54-sample test set for SVM (left) and kNN (right).

4.4 ROC Curves

One-vs-rest ROC curves were plotted for each class, for both classifiers. The Area Under the Curve (AUC) for each class is shown in the legend.

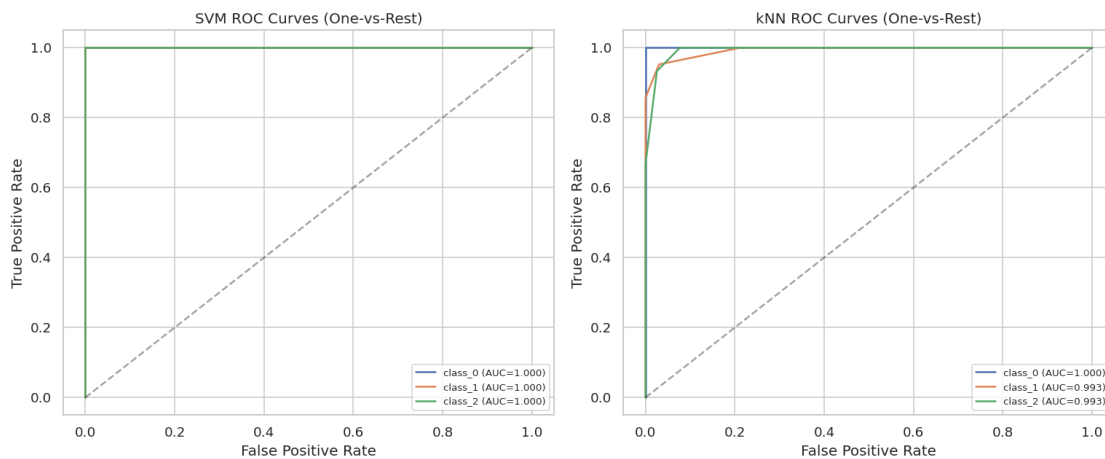


Figure 4.3: ROC curves (one-vs-rest) for SVM (left) and kNN (right).

4.5 Discussion and Inferences

- **SVM outperforms kNN on all three metrics:** accuracy (0.981 vs 0.944), kappa (0.972 vs 0.917), and ROC-AUC (1.000 vs 0.995). This is consistent with the dataset characteristics observed in Task A: the classes are well-separated but the separation involves a combination of multiple features rather than simple axis-aligned boundaries, which the RBF kernel's non-linear decision boundary captures more effectively than kNN's purely local, distance-based voting.
- **SVM achieved a perfect ROC-AUC of 1.0**, meaning that for every pairwise one-vs-rest comparison, there exists some threshold at which the SVM perfectly separates that class from the rest – even though one sample was still misclassified in the confusion matrix (a class_2 sample predicted as class_1). This shows AUC and accuracy capture different things: AUC reflects the ranking quality of predicted probabilities across all thresholds, not just the single threshold (0.5) used to produce the confusion matrix.
- **kNN's main confusion is between class_1 and class_2** (3 samples of class_1 misclassified as class_2). Referring back to the boxplots in Figure 3.2, class_1 and class_2 have the most overlapping flavanoid and color_intensity ranges among the three classes, which likely causes kNN's nearest-neighbour votes to occasionally favour the wrong class near the decision boundary.
- **Cohen's Kappa above 0.9 for both models** confirms that both classifiers are performing far better than random guessing (which would give kappa ≈ 0), and the gap between SVM's kappa (0.972) and kNN's (0.917) reflects SVM's more consistent performance specifically on the harder-to-separate

class_1/class_2 boundary.

- **Practical takeaway:** for this dataset, a kernel-based method that can model curved decision boundaries (SVM) is preferable to a purely distance-based method (kNN), though both achieve strong (>90%) accuracy, confirming that the 13 chemical features collectively carry strong discriminative signal for the three cultivars.

5. Task C — Clustering (k-Means and DBSCAN)

5.1 Purpose and Methodology

Unlike Task B, clustering is **unsupervised**: the true class labels are not used to fit the model. The goal is to test whether the natural groupings (clusters) discovered purely from the 13 chemical features correspond to the three known wine cultivars. Two algorithms were compared:

- **k-Means (k=3)** – partitions the data into exactly 3 clusters by iteratively assigning points to the nearest centroid and recomputing centroids. k=3 was chosen to match the known number of classes, and validated using the elbow method (Figure 5.1).
- **DBSCAN** – a density-based algorithm that groups together points that are closely packed, marking points in low-density regions as *noise* (outliers). It does not require the number of clusters to be specified in advance. Parameters used: $\text{eps}=2.2$, $\text{min_samples}=3$, selected using the k-distance plot (Figure 5.2).

Two metrics were used to evaluate and compare the clusterings:

- **SSE (Sum of Squared Errors / Inertia)** – the total squared distance of each point from its cluster centre. Lower SSE means tighter, more compact clusters.
- **Entropy** – for each cluster, the entropy of the true class labels within that cluster is computed (in bits), then averaged across clusters weighted by cluster size. Lower entropy means each cluster predominantly contains samples from a single true class (i.e., the unsupervised clusters align well with the real cultivars).

5.2 Choosing k for k-Means: Elbow Method

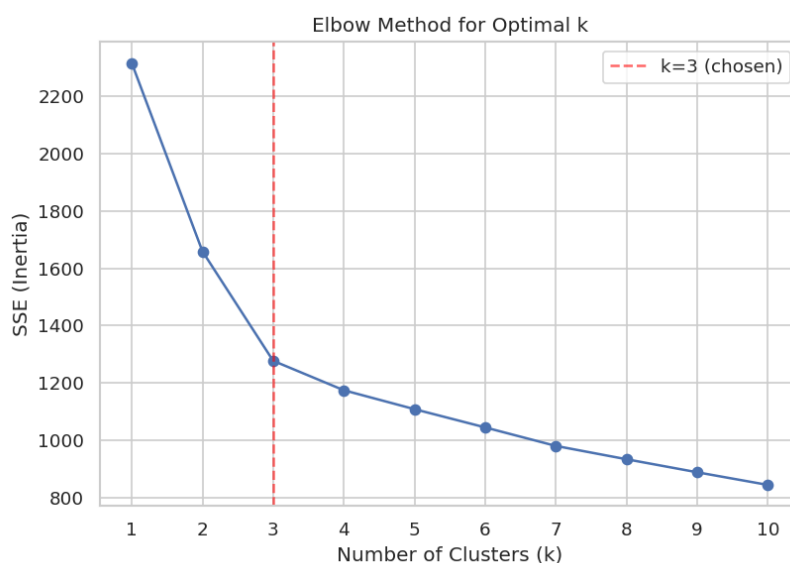


Figure 5.1: SSE vs number of clusters (k). The 'elbow' near k=3 supports using 3 clusters, matching the 3 known wine classes.

5.3 Choosing eps for DBSCAN: k-Distance Plot

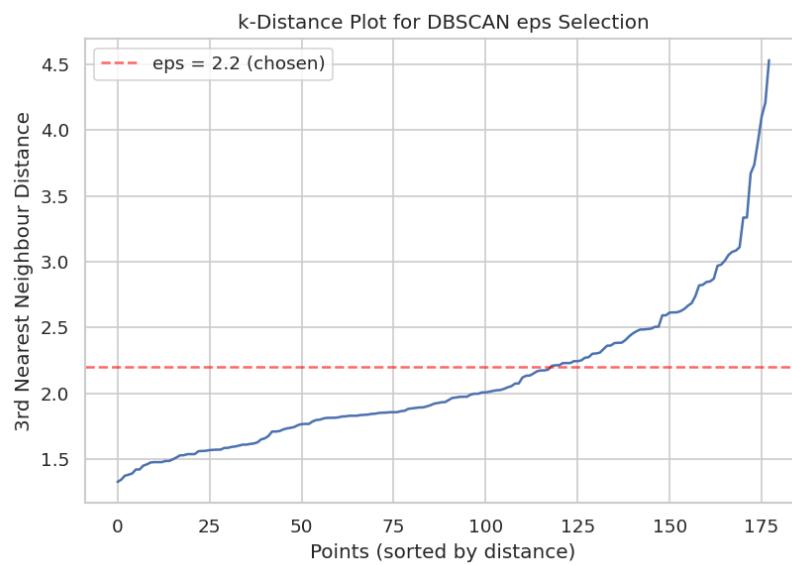


Figure 5.2: 3rd-nearest-neighbour distance for each point, sorted ascending. The 'knee' of this curve guides the choice of eps; eps=2.2 was chosen to balance cluster count against noise points.

5.4 Results Comparison

Metric	k-Means (k=3)	DBSCAN (eps=2.2, min_samples=3)
SSE	1277.93	965.69
Entropy (bits)	0.1898	0.8057
Silhouette Score	0.2849	0.0725
Clusters Found	3 (fixed)	3
Noise Points	0 (n/a)	44

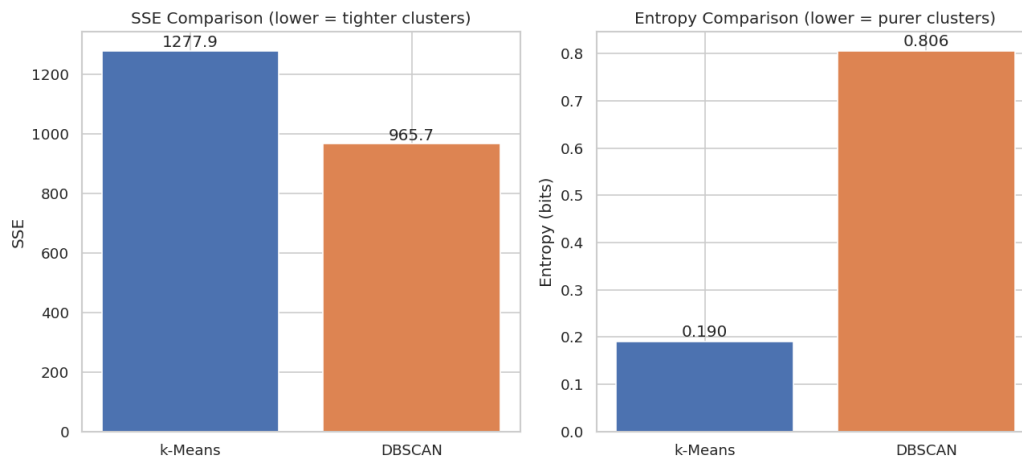


Figure 5.3: SSE (left) and Entropy (right) for k-Means vs DBSCAN.

5.5 Visualizing the Clusters (PCA Projection)

Since the data has 13 dimensions, Principal Component Analysis (PCA) was used to project it down to 2 dimensions (capturing the two directions of greatest variance) purely for visualization. The true class labels, k-Means clusters, and DBSCAN clusters are plotted side-by-side on the same 2D projection.

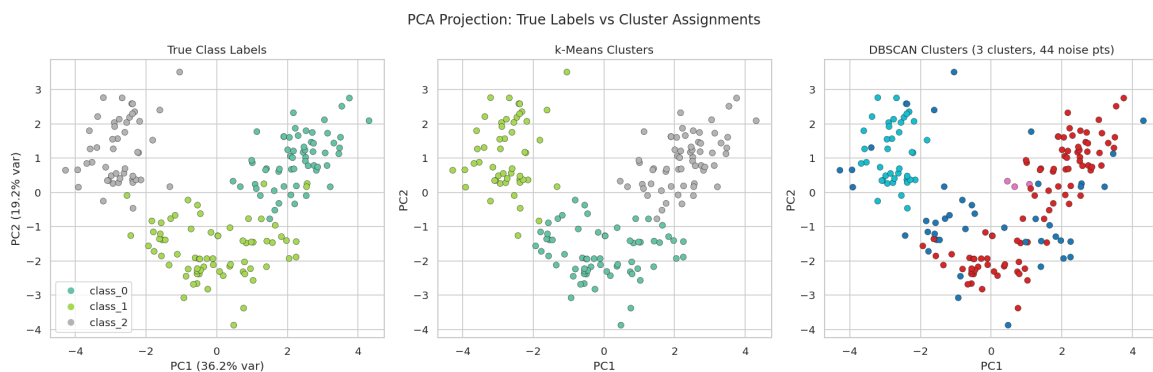


Figure 5.4: PCA projection (PC1 vs PC2) showing true class labels (left), k-Means clusters (centre), and DBSCAN clusters with noise points in dark blue (right).

5.6 Discussion and Inferences

- **k-Means produces purer clusters (lower entropy):** with $k=3$, k-Means achieved an entropy of only 0.190 bits, meaning each k-Means cluster is dominated almost entirely by a single true wine class. Comparing the left and centre panels of Figure 5.4, the k-Means clusters visually overlay the true class regions very closely – this confirms that the three cultivars form genuinely distinct, roughly spherical/convex groups in the standardized feature space, which is exactly the kind of structure k-Means is designed to find.
- **DBSCAN achieves lower SSE** (965.7 vs 1277.9 for k-Means) **but much higher entropy** (0.806 vs 0.190 bits). The lower SSE is partly an artifact of DBSCAN excluding 44 boundary/outlier points as 'noise' – removing these points naturally shrinks the within-cluster spread of the remaining points. However, the much higher entropy shows that the clusters DBSCAN *does* find mix samples from multiple true classes more than k-Means does.
- **Why DBSCAN struggles here:** DBSCAN assumes clusters are separated by regions of *low density*, and works best when clusters have roughly uniform density and clear density 'gaps' between them. Looking at Figure 5.4 (right panel), the three wine classes in PCA space form a connected, continuously-varying region rather than being separated by sparse gaps – especially class_0 and class_2, which sit close together. As a result, DBSCAN with a single global eps value either merges multiple true classes into one density-connected region, or splits a single true class across multiple density-based clusters, depending on local density variation.
- **Trade-off between SSE and Entropy:** This comparison illustrates an important general point – SSE alone is not a reliable indicator of 'good' clustering when ground-truth labels are available for validation. DBSCAN's lower SSE looks favourable in isolation, but the entropy metric (which uses the true labels) reveals that k-Means' clusters are far more *meaningful* with respect to the actual wine cultivars.
- **Practical takeaway:** for datasets where the true groups are convex, roughly equal-sized, and globular – as the three wine cultivars appear to be in this 13-dimensional standardized space – centroid-based methods like k-Means recover the true structure more faithfully than density-based methods like DBSCAN, which are better suited to clusters of irregular shape with clear density gaps between them.

6. Overall Conclusions

- The Wine Recognition dataset (178 samples, 13 numeric chemical features, 3 balanced classes) required minimal cleaning – no missing values were present – but **standardization was essential** due to the widely differing scales of the 13 features.
- Correlation analysis and class-wise boxplots showed that features such as **flavanoids**, **proline**, and **color_intensity** are strong natural discriminators between the three cultivars, explaining the high downstream classification accuracy.
- For **classification**, **SVM (RBF kernel) outperformed kNN** on accuracy, Cohen's Kappa, and ROC-AUC, achieving near-perfect separation of the three classes (98.1% accuracy, kappa 0.972, AUC 1.000).
- For **clustering**, **k-Means (k=3) produced clusters that align much more closely with the true cultivars** (entropy 0.190 bits) compared to DBSCAN (entropy 0.806 bits), even though DBSCAN achieved a lower raw SSE by treating 44 points as noise. This highlights that no single metric (SSE) should be used in isolation to judge clustering quality.
- Overall, the analysis confirms that the 13 chemical measurements collectively form a feature space in which the three wine cultivars are **well-separated and largely convex/globular** – a structure well-suited to both kernel-based classifiers (SVM) and centroid-based clustering (k-Means), but less suited to density-based clustering (DBSCAN) at a single global density threshold.

All code, plots, and this report were generated using Python (pandas, NumPy, scikit-learn, matplotlib, seaborn, reportlab). The complete source code is available in the accompanying repository.